2025 5th International Conference on Sensors and Information Technology

An Autonomous Charging System for A Two Wheeled Movable Robot

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***Abstract*—This paper designs an autonomous charging system for a two-wheeled mobile robot. The system utilizes LiDAR to identify a charging station with a specific structure, collects and processes data to determine the target point, and employs a feedback control law to achieve smooth motion of the robot. Ultimately, the robot precisely docks with the charging station and total the charging job.**

Keywords-autonomous recognition; laser radar; two wheeled movable robot; charging post

I. INTRODUCTION

The development of chic farming in China has got high thinking from the administration, which has introduced a sequence of procedure to help the request of technology in agriculture and creature farming. These policies aim to address the problems of low quality and efficiency, inadequate support and guarantee systems, and weak risk resistance in the development of animal husbandry. In 2021, the 14th Five Year Plan highlight the improvement of farming standard and effectiveness, as well as the improvement of agricultural infrastructure and environmental surroundings. In the same year, the Cyberspace Administration of China and ten other departments released the "Action Plan for Digital Rural Development (2022-2025)", put forward the improve of digital infrastructure and change in chic agriculture to help the digital and clever change of farming.[1]

With the rapid development of animal husbandry in China, mobile robots are gradually being applied in this field. This type of robot relies on its own battery to provide kinetic energy. When the battery level is below the critical value, it needs to be connected to a power source for charging. Therefore, achieving autonomous charging has become a key technology for mobile robots to continue working[2].

Lidar positioning technology, as the mainstream means of mobile robot navigation and positioning, its positioning accuracy is not yet sufficient to ensure the foolproof charging process. At present, there are various technological approaches for mobile robots to achieve autonomous charging, including but not limited to device or track guidance, infrared array pointing, and laser scanning of surrounding environmental features [3-5]. These technical solutions often rely on additional hardware components, complex technical support, or detailed environmental parameter mapping.

Path arrangement is also one of the core job in movable robot sailing, aiming to discover an optimal or practicable way from the starting point to the mark essence in a given surroundings. Classical methods include the A* algorithm, Dijkstra algorithm, and others. The A* algorithm has been widely used in static environments, but its computational complexity increases significantly in high-dimensional or dynamic environments. Additionally, the paths generated by A* may lack smoothness, leading to jittery or unstable motion when the robot executes the path. To address the issue of path smoothness, researchers have proposed various improvement methods. For example, RRT and its variants such as RRT*, path smoothing methods based on MPC feedback control laws, and the introduction of reinforcement learning techniques have been developed[6-9]. Although these approaches can generate smooth and dynamically adjustable paths, the processes are complex and computationally expensive, necessitating further optimization.

This article proposes a charging device design with feature identification, and based on the recognition of feature data of the device and navigation control based on feedback control law, the robot can accurately locate and smoothly move to the charging device to complete autonomous charging tasks.

II. STRUCTURE AND POSE

A. Structure

1) Structure of Movable Robots:

Install laser radar, energy storage battery system, motion control system, wireless communication system, and wireless charger power receiving end on the robot.

The laser radar adopts a single line radar form and is horizontally installed on the robot. The strength storage battery system consists of an energy storage battery and a position feedback module, where the status feedback module can supply real-time facts on the left-over battery position and charging/ release status. The movement control system is responsible for adapt the present of the robot, containing left-hand and right revolution and leading and reverse group. Wireless communication devices are used to achieve communication between robots and charging devices. The power receiving end of the wireless charger serves as the input interface for the robot's charging function, used to receive power input from the charging device's transmission end, as shown in Figure 1.

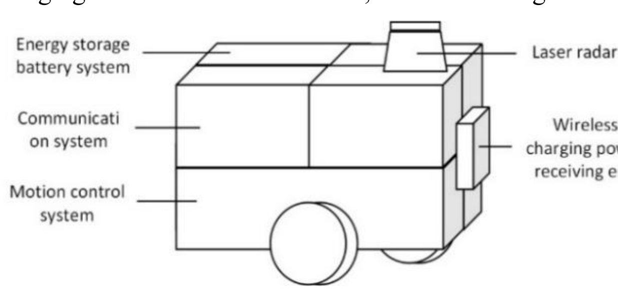

Figure 1. The fundamental five pieces arrangement of movable robots

2025 5th International Conference on Sensors and Information Technology (ICSI) | 979-8-3315-4288-7/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICSI64877.2025.11009700

2) Structure of charging device

The charging device includes a charging station and a charging control device. The charging station includes a wireless charger transmission end, a characteristic structure, and a power conversion module; The charging control device includes a control module and a wireless communication module, as shown in Figure 2.

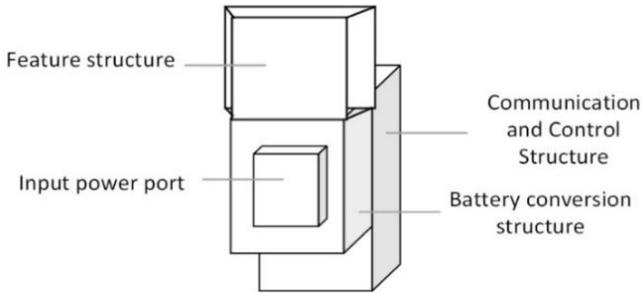

Figure 2. The basic four parts structure of the charging pile

3) The relationship between robot structure and charging pile structure

The power get end of the robot is installed in head of the robot, at the same tallness as the control transfer close of the charging heap. The LiDAR is installed directly above the front of the robot, within the scanning plane and the characteristic structure range of the charging device.

B. *The pose of the robot*

This article considers using two independently driven parallel wheels to implement the robot described in the two wheeled vehicle model, so that the linear velocity and angular velocity can be independently controlled. We use a freezing organize system to relate the movement of machine.

A LiDAR sensor is located on the robot and focuses on the target q at a distance of r from the robot. Let $\emptyset \in (-\pi, \pi]$ be the way of q relation to the stroke of vision. Let $\alpha \in (-\pi, \pi]$ be the gradient between the robot's title and the robot's leading way, as indicate in Figure 3.

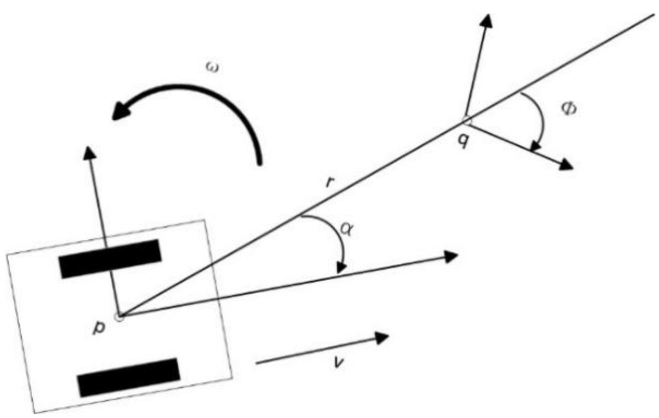

Figure 3. Contrast to the freezing organize model accepted with the movable robot p as the center, q is the recognize mark essence, and r is the direct - stroke space between the two.

So, it's easy to see that the kinematics of the robot can be written as formula (1).

$$\begin{pmatrix} r \\ \emptyset \\ \alpha \end{pmatrix} = \begin{pmatrix} -v\cos\alpha \\ \frac{v}{r}\sin\alpha \\ \frac{v}{r}\sin\alpha + \omega \end{pmatrix} \tag{1}$$

We design a control system that allows the robot to reach the target position stably. According to alpha (robot heading), its function is to adjust the robot's heading, gradually bringing the robot's position closer to the target point (origin). It can be imagined as' telling the robot which direction to go '. Afterwards, swiftly adapt the robot to make certain that its α encounter the necessity of practical control and accomplish the intended robot present. Eventually, transfer the vehicle to the mark location.

III. IDENTIFICATION AND CONTROL

A. *Control*

According to the movement model mentioned earlier, the target essence is resolute through LiDAR and the machine is power to transfer to the mark essence. Transform this problem into the process of taking (r, Ø, α) to the origin based on a given initial posture.

Here we quote a feedback control law that dynamically adjusts the control input by measuring the deviation between the system output and the expected state, stabilizing the system in the expected state and achieving stable control of the system. Let x act for the dissimilarity between the genuine machine title and the anticipated value arctan ($-k_1 \emptyset$), as state in formula (2).

$$x = \alpha - \arctan(-k_1\emptyset) \tag{2}$$

To analyze the stability of this control constitution, the Lyapunov firmness hypothesis can be working. For the given control law x, the following Lyapunov function can be constructed:

$$V(\phi) = \frac{1}{2}\phi^2 \tag{3}$$

This purpose content the optimistic definiteness calculate because $V(\phi) > 0$ for all $\phi \neq 0$.

To check the firmness of the system, it is necessary to compute the unoriginal of $V(\phi)$.

$$\dot{V}(\phi) = \phi\dot{\phi} \tag{4}$$

Presume the energetic equating of the system is as follows:

$$\dot{\phi} = -k_1 \frac{-k_1}{1 + (k_1\emptyset)^2}\dot{\phi} \tag{5}$$

Substituting into the system yields:

$$\dot{V}(\phi) = \phi\left(\frac{{k_1}^2}{1 + (k_1\emptyset)^2}\dot{\phi}\right) \tag{6}$$

Moreover, the sign of ϕ is opposite to that of $\dot{\phi}$, thus: $\dot{V}(\phi) < 0$. This indicates that the system is asymptotically stable at the equilibrium point $\phi = 0$. This demonstrates that the control law possesses Lyapunov stability, making certain the firmness and meeting of the system.

Therefore, by amalgamate (1), it is simple to get the unoriginal of x and its association with ω, as follows:

$$\dot{x} = \left(1 + \frac{k_1}{1 + (k_1\phi)^2}\right)\frac{v}{r} sin(x + arctan(-k_1\phi)) + \omega \quad (7)$$

After sorting, the formula for ω is obtained:

$$\omega = -\frac{v}{r}\left[k_2\big(\alpha - \arctan(-k_1\emptyset)\big) + sin\alpha + \frac{k_1 sin\alpha}{1 + (k_1\emptyset)^2}\right] \quad (8)$$

Where $k_1>0$ and $k_2>0$ are used to control non-zero positive gain. In planar movement, there is a near association between skinny velocity, linear speed, and curving. The curvature $\boldsymbol{\lambda}$ can be expressed as ω / v. Therefore, the curvature $\boldsymbol{\lambda}$ can be record as formula (9).

$$\lambda = -\frac{1}{r}\left[k_2\big(\alpha - \arctan(-k_1\emptyset)\big) + sin\alpha + \frac{k_1 sin\alpha}{1 + (k_1\emptyset)^2}\right] \quad (9)$$

The process of representing the path is not affected by v, and ultimately the vehicle can move smoothly to the target posture.

B. *Identification*

In the autonomous charging system based on LiDAR, this paper adopts the Iterative Closest Point (ICP) algorithm to achieve precise positioning and alignment of the robot relative to the charging station. Thus determining the pose of the robot. Its mathematical model can be expressed as:

$$\min_{R,t} \sum_{i=1}^{N} \| Rp_i + t - q_j \|^2 \quad (10)$$

Where R is the revolution matrix (R ∈ $R^{2\times2}$). T is a interpretation vector (t ∈ R^2). p_i is the point in the flow essence mist. q_j is the near essence in the map point mist corresponding to p_i.

Particularly, first roughly align the two sets of essence clusters, and then continuously adjust their poses to find their corresponding relationship. At each adjustment, the average essence cloud error (RMSD) between the two should be calculated. If the calculation result is less than the set value or the number of adjustments reaches a certain value, the calculation ends and the final corresponding relationship is obtained.

$$\text{RMSD} = \sqrt{\frac{1}{N}\sum\nolimits_{N}^{i=1} \| p_i - q_i \|^2} \quad (11)$$

When the mistake becomes sufficiently small, drop below a predefined entrance, or when the greatest numeral of iterations is reached, the algorithm end and production the last change consequence. This method improves positioning accuracy and ensures precise docking during the charging process.

C. *Empirical test*

Before actual operation, perform computer simulation work. This feature uses VMware Workstation Pro 16 practical machine to set up Ubuntu 22.04 operating system in the facility appliance simulation test software, and configures the Humble version of ROS2 environment. The movable machine uses SLAM to produce maps and uses ROS2 (robot operating system) navigation pile for self-ruling sailing.

When the energy storage battery system of the mobile robot indicates that the battery level is low, and the robot is far from the charging station in the Gazebo simulation environment, it is necessary to navigate the robot within a 1-meter range of the charging station. The robot's motion process during this period is shown in Figures 4 to 6.

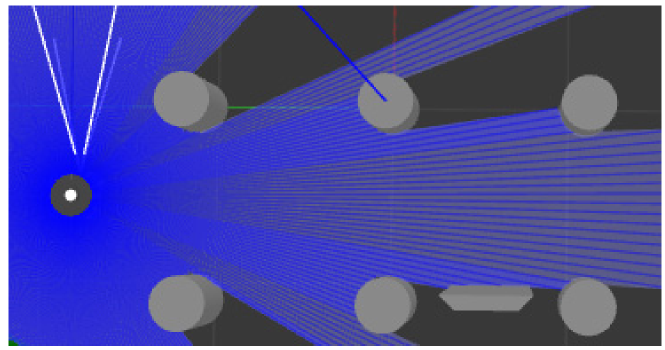

Figure 4. The robot is more than one meter away from the charging station. One grid on the map represents 1 meter

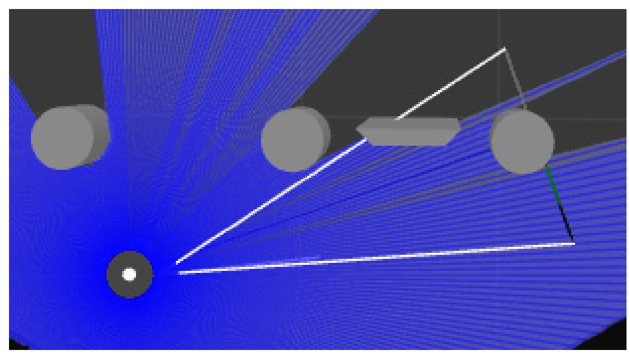

Figure 5. The process of robot navigation moving to the charging station

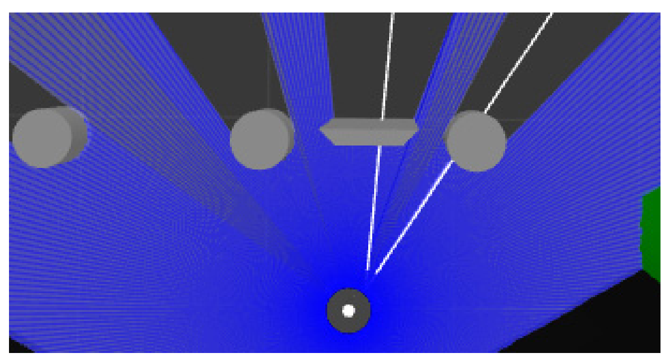

Figure 6. The robot navigates to the vicinity of the charging station and begins to recognize and autonomously dock

When the mobile robot moves within a 1-meter range of the charging station with a special structure, the robot's position in the simulation environment matches its position on the map. At this point, the robot's position is considered as the initial pose.

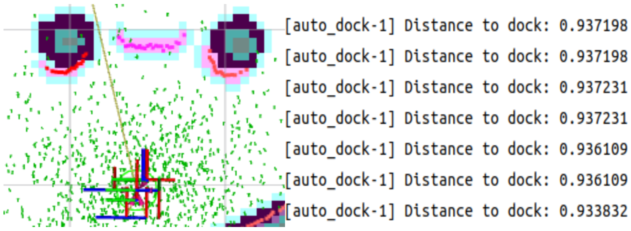

Figure 7. The left side represents the relationship between the initial pose of the robot and the target point in Rviz 2, while the right side represents the real-time distance recognized between the two during autonomous navigation

Set the opening location of the robot in the copy surroundings to a space of almost r=1.0m from the exceptional arrangement. As a basis, corresponding constraints are set, such as angular velocity and linear velocity, to ensure that the robot not only moves to the target point with ease, but also stops steadily when it reaches a certain position. The relevant value is $v_{max} = 0.5\, m/s$,$v_{min} = 0.1\, m/s$, $\omega_{max} = 1.0\, rad/s$

By inputting the command line through the ROS2 terminal, the robot model performs autonomous navigation docking. Using the Rviz2 visualization tool, the constructed service robot model can be viewed in real time. The initial distance between the model and the special structure observed at the output end of the terminal is 0.937198 meters, and then the distance gradually decreases. A distance less than 0.3 meters indicates successful docking.

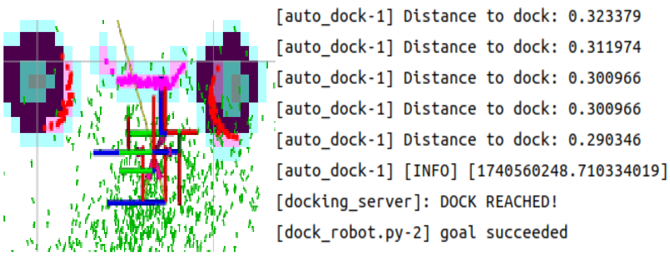

Figure 8. The left side represents the map of successful docking, while the right side shows the data and text indicating successful docking output from the terminal

Based on the experimental process and results shown in Figure 7 to 8 above, the proposed method enables mobile robots to quickly adjust appropriate poses, smoothly move to target points, accurately dock, and complete autonomous docking tasks.

IV. CONCLUSIONS

With the significant increase in the number of smart agriculture in China, the demand for mobile robots has also greatly increased. Through this study, a simulated map of environmental information was constructed using laser radar sensors for charging recognition. The charging process was coordinated and controlled on the ROS2 system, and a complex path planning algorithm was validated to build an autonomous charging system for a robot based on laser radar sensors. Smooth motion control ensures the safety of the autonomous charging system during the charging process, providing an efficient energy supply solution for smart agricultural mobile robots.

ACKNOWLEDGMENT

We sincerely thank the Jinan Science and Technology Program "Leader Open Bidding" project (202428071), the Qilu University of Technology's (Shandong Academy of Sciences) Scientific Major Innovation Project "Bidding System" project (2023JBZ01), and the 4000T Automotive Beamforming Multi-Robot Cooperative Manufacturing Automation Equipment Research and Demonstration Application Project (2022TSGC1062) for the support and guidance of this paper.

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